EC39001: Analog Communication Lab
Experiment 7: Sampling

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1 Aim of the Experiment

The objectives of this experiment are:

1. To sample and hold the given input signal.
2. To generate a Pulse Amplitude Modulated (PAM) signal and to Demodulate it.

2 Components Used

- LM 741 Operational Amplifier
- JFET (2N3819 or equivalent)
- Resistors: $1k\Omega$, $10k\Omega$, $47k\Omega$, etc.
- Capacitors: $6.8nF$, $100nF$
- Function Generator
- Mixed Signal Oscilloscope (DSO)
- Breadboard and connecting wires
- $\pm 12V$ DC Power Supply

3 Theory & Circuit Diagram

3.1 Sample and Hold Circuit

A **sample and hold (S/H) circuit** is used to sample the instantaneous value of an analog signal and hold it for a certain duration. This is important in data conversion systems such as ADCs.

Key Parameters:

- **Aperture Time:** The time during which the sampling occurs. It should be as short as possible for accurate capture.
- **Hold Time:** The duration for which the sampled value is held constant by the holding capacitor.

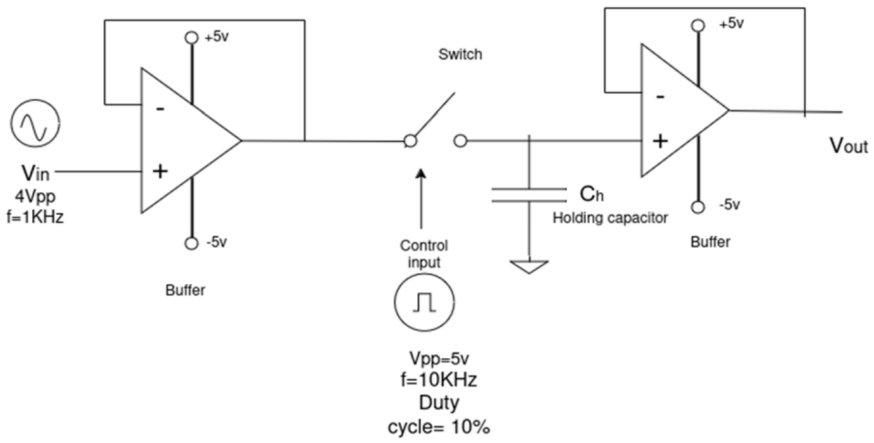


Figure 1: Sample and Hold Circuit Diagram

3.2 Pulse Amplitude Modulation (PAM)

Pulse Amplitude Modulation is a technique where the amplitude of a train of pulses varies according to the instantaneous value of the modulating signal.

It is an essential step in analog-to-digital conversion and pulse communication systems. A JFET can act as an analog switch to control the sampling of the input signal based on a pulse waveform.

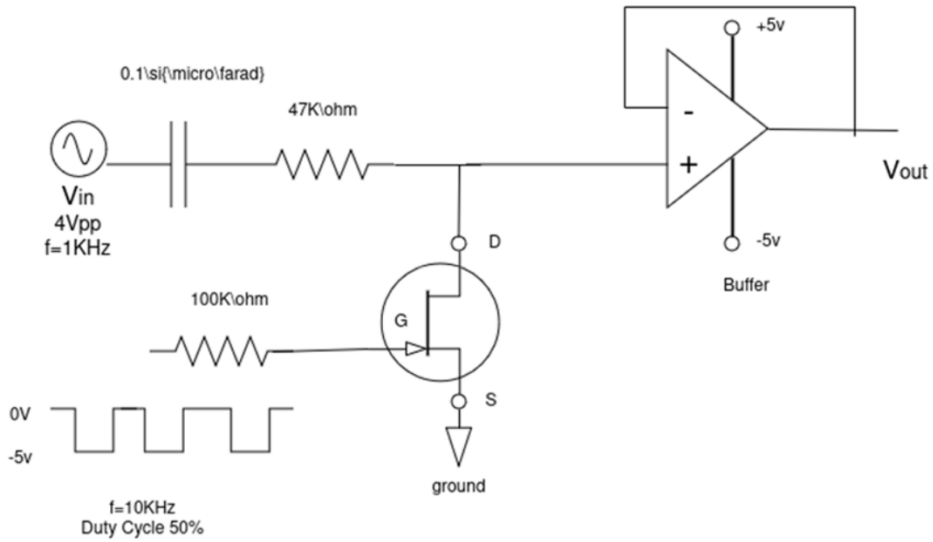


Figure 2: PAM Circuit Diagram

4 Observations and Discussion

4.1 Sample and Hold Operation

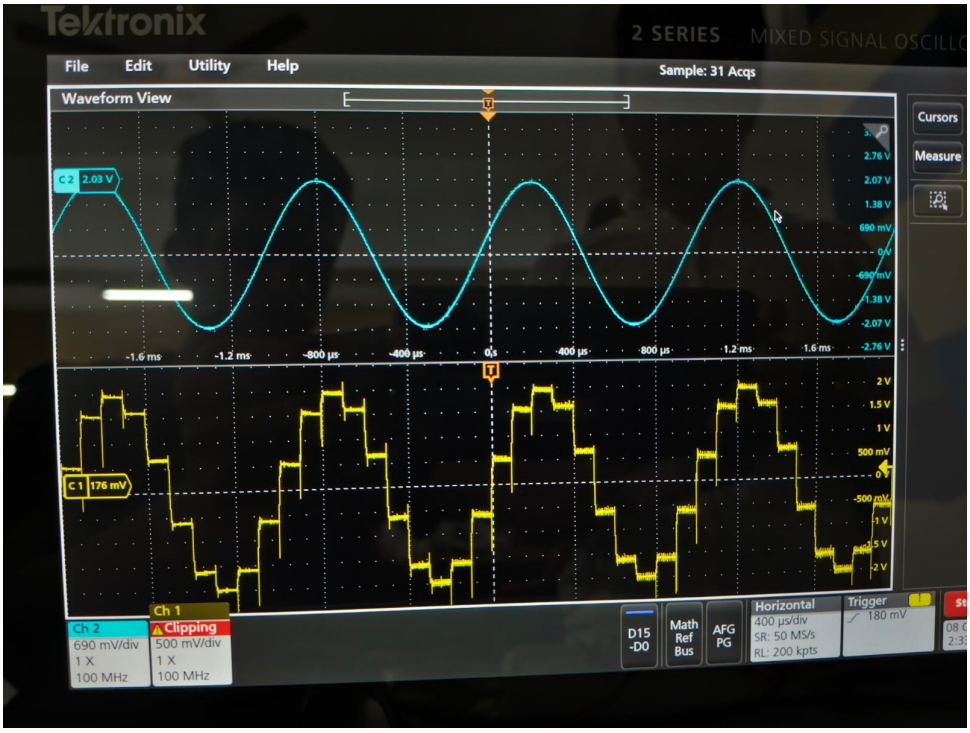


Figure 3: Sample and Hold Output for 10% Duty Cycle and 6.8nF Capacitor

In Figure: 3 we can observe output of sample and hold circuit ,which is as expected . There are some high frequency noise components.

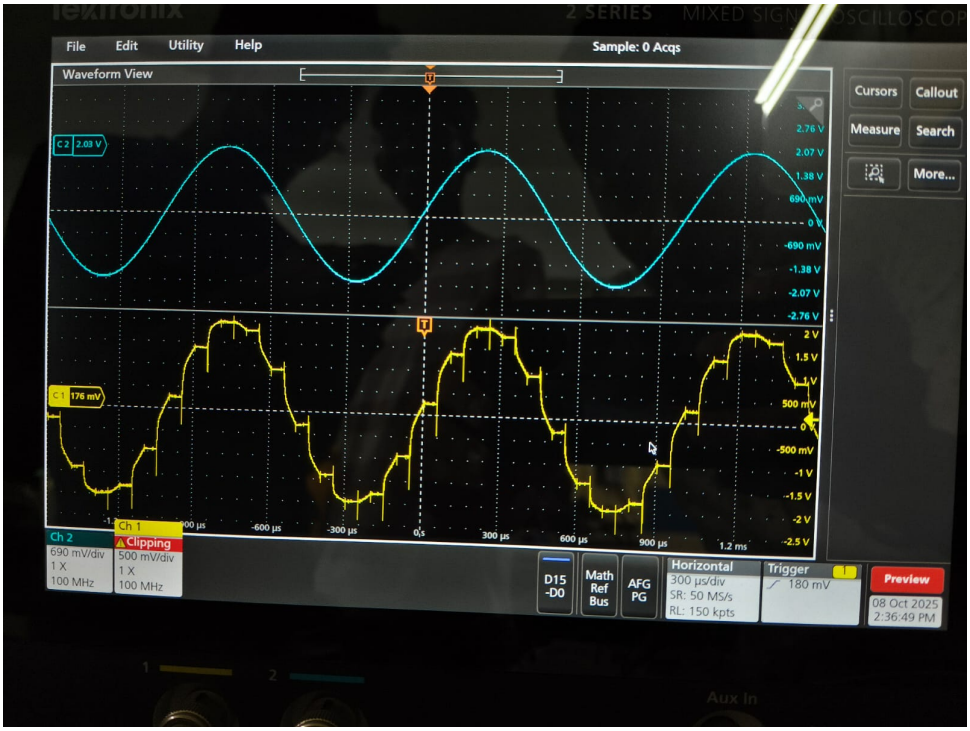


Figure 4: Sample and Hold Output for 50% Duty Cycle and 6.8nF Capacitor

In Figure 4 we have used duty cycle of square wave to be 50 % duty cycle. As a result we can see the 50 % of the message signal as it is and other 50 % part it is flat DC value(hold).

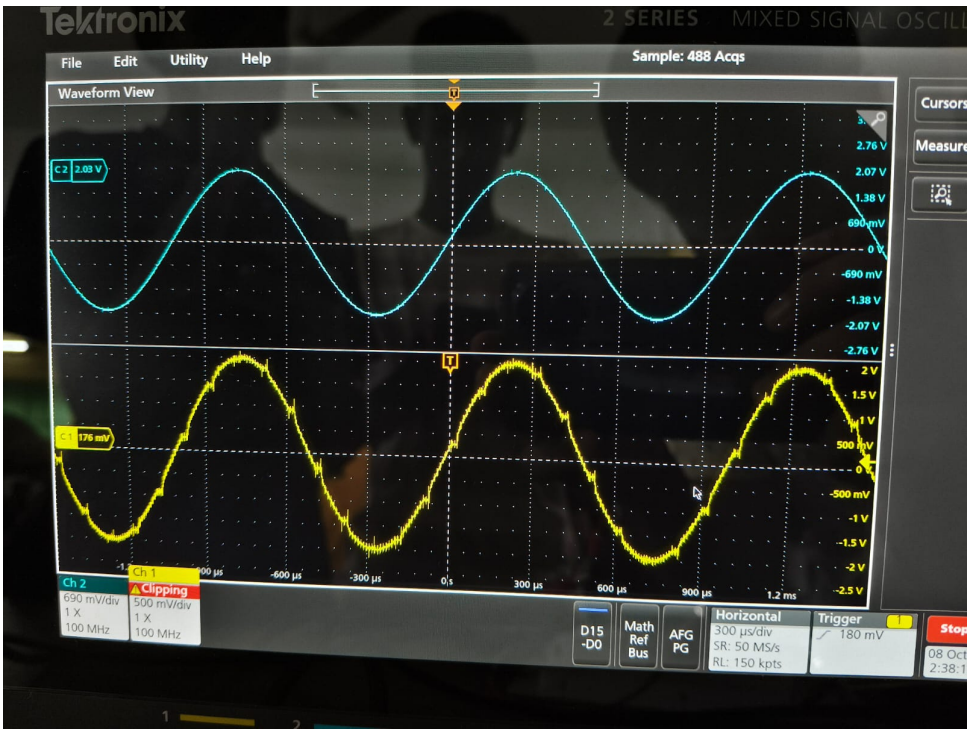


Figure 5: Sample and Hold Output for 80% Duty Cycle and 6.8nF Capacitor

In Figure 5 we have used 80 % duty cycle as a result we can observe that output

is almost similar to original signal with slight holding part where output is flat(20 % of time).

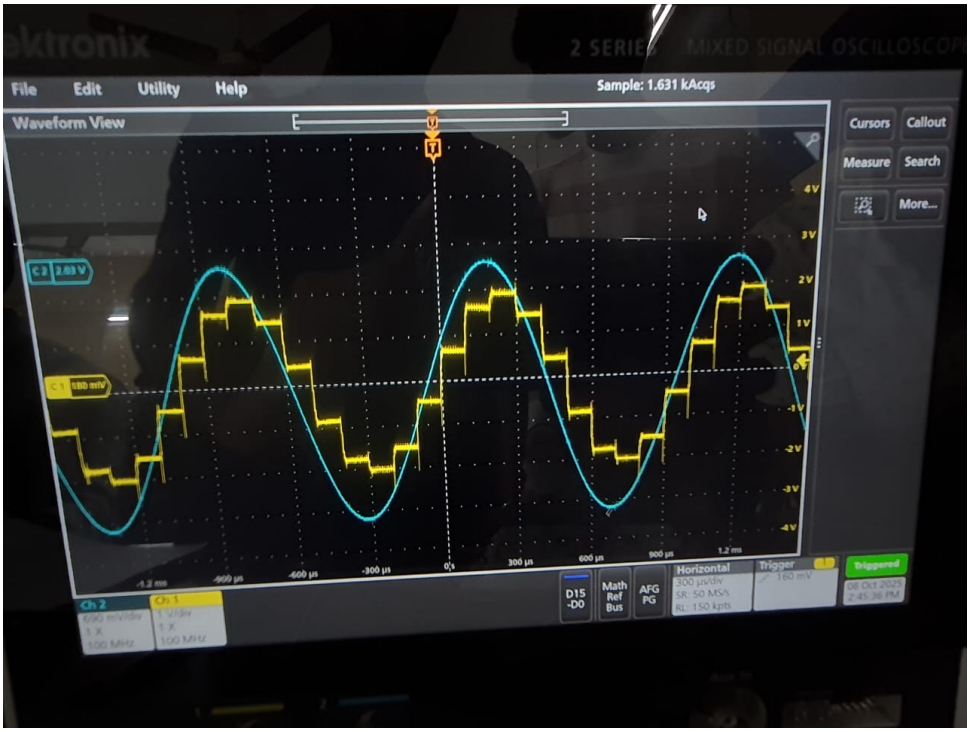


Figure 6: Sample and Hold Output for 15% Duty Cycle and 6.8nF Capacitor

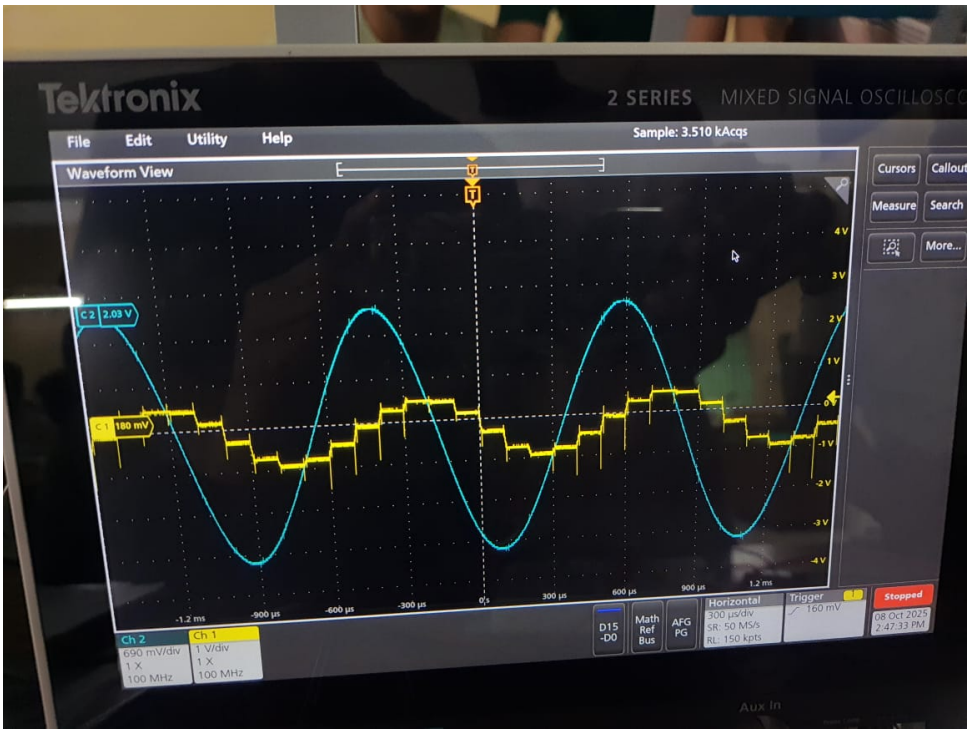


Figure 7: Sample and Hold Output for 15% Duty Cycle and 100nF Capacitor

In Figure 6 and 7 we have compared the effect of capacitor value at equal duty cycle of square wave. In Fig:7 we can observe that due to high cap value its charging time is

higher and hence not able to catch up with the signal,as we can observe the difference between two signals.

4.2 Pulse Amplitude Modulation

The PAM waveform was obtained by applying a sinusoidal input and controlling the JFET switch using pulse signals. The amplitude of each pulse follows the instantaneous value of the input sine wave.

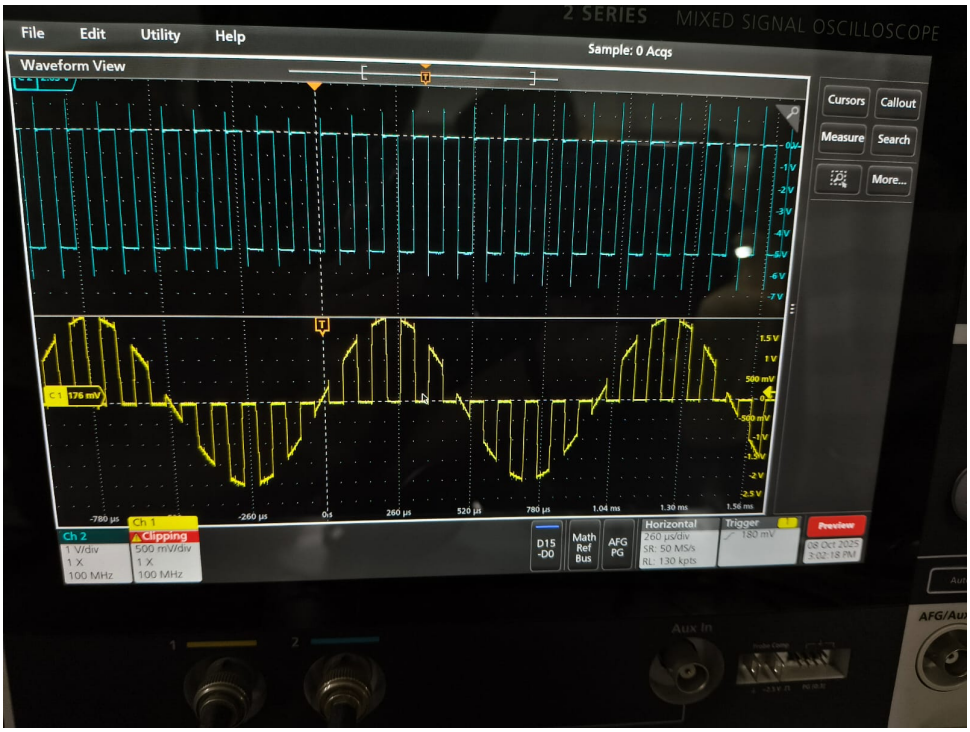


Figure 8: Output of Pulse Amplitude Modulated signal

In Fig 8 we can see the clean output of PAM signal ,which is as expected.

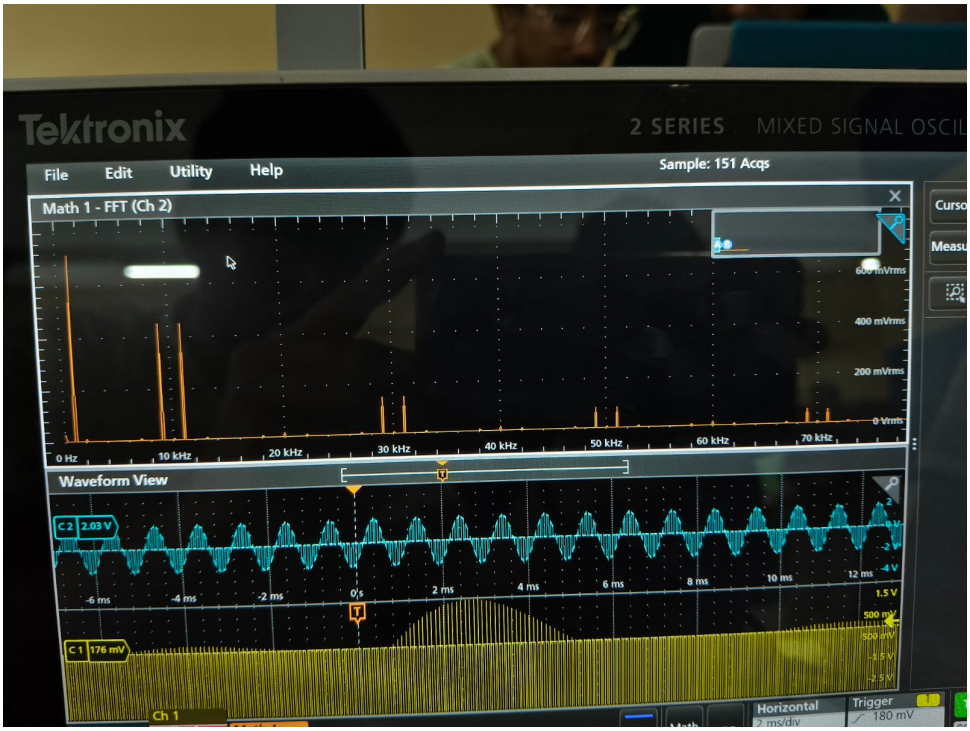


Figure 9: FFT of Output of Pulse Amplitude Modulated signal

In Fig 9 we can see pair of peaks both sides(-1K and +1K) of odd hormonics(0 10k , 30k , 50k). This can be easily be understand as our square wave is of 10kHz frequency and our modulating signal is of 1kHz.

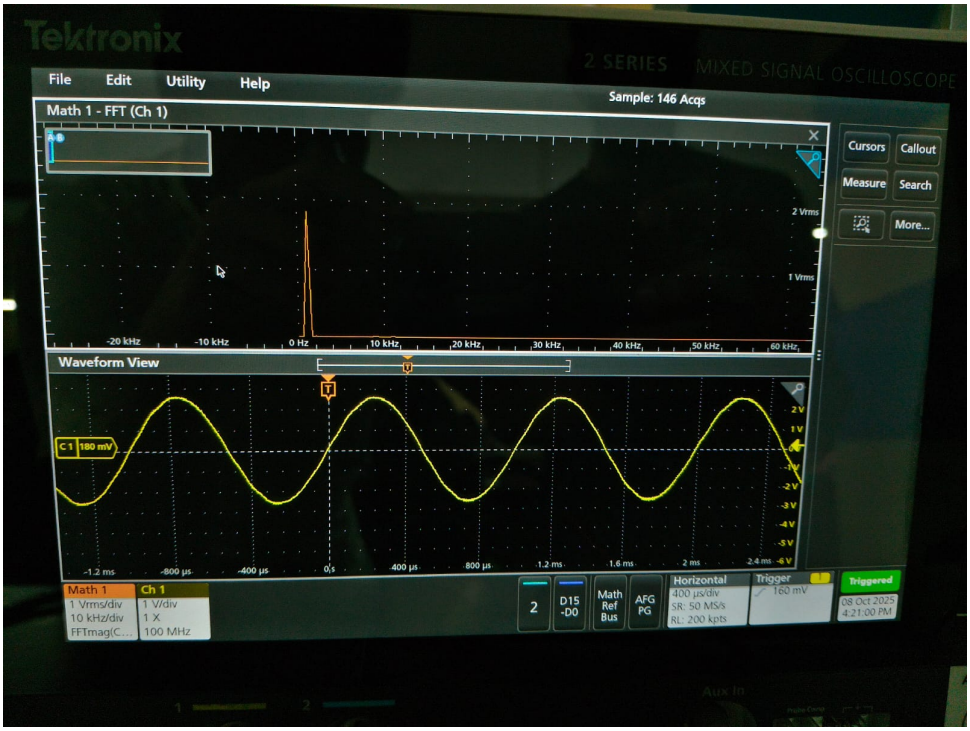


Figure 10: Demodulated PAM signal

In Fig 10 we have applied an active 2nd order low pass filter with cut off frequency

2kHz to isolate our modulating signal which is of 1kHz frequency. As we can observe in the fig that we have succesfully recovered our message signal with minimum noise.

Discussion

- In this experiment, we successfully performed **Sample and Hold**, **Pulse Amplitude Modulation (PAM)**, and **demodulation of the PAM signal**.
- The sample and hold circuit accurately captured and maintained the instantaneous value of the input waveform during the hold period.
- As the duty cycle of the control pulse increased from 10
- However, for precise sampling, the pulse width should be kept as small as possible to avoid distortion and overcharging effects.
- The effect of capacitance on the sampling accuracy was also studied. For 10 kHz sampling frequency (10 µs ON time), a 6.8 nF capacitor with a 1.7 µs charge time gave proper results.
- Using a larger 100 nF capacitor caused a charge time of about 25 µs, leading to incomplete charging and incorrect output levels.
- The ON resistance of the switching modulator (JFET) varied between 200–650 Ω, which influenced the charging rate of the holding capacitor.
- In the PAM circuit, a 10 kHz square wave was used as the carrier and a 1 kHz sine wave as the modulating signal.
- The obtained PAM waveform showed amplitude variations corresponding to the input signal, confirming correct modulation.
- The FFT spectrum of the PAM output showed peaks at ±1 kHz around the carrier and its odd harmonics (10 kHz, 30 kHz, etc.), along with a DC component near 0 Hz due to the square carrier.
- The original 1 kHz signal was successfully recovered using a 2nd-order active low-pass filter with a cutoff frequency around 2 kHz.

——END OF REPORT——